

Problem Definition: Student Management System (CRUD-Based)

Objective:

Design and implement a **Student Management System** that allows users to **Create**, **Read**, **Update**, and **Delete** student records using a **menu-driven Java application**.

Student Data Fields:

Each student will have the following attributes:

Field	Type	Description
Roll No.	Integer	Unique identifier (Seat No)
Name	String	Full name of student
Gender	String	"Male", "Female", or "Other"
Contact	String	Mobile number
Year	Integer	Academic year (1 to 4)
Branch	String	Department (e.g., CSE, IT)

Operations to Implement (CRUD)

Operation	Description
Create	Add a new student record
Read All	Display all student records
Search	Search student by Roll No.
Update	Update student record using Roll No.
Delete	Delete a student record using Roll No.

----- Student Management System -----

1. Add Student
2. Display All Students
3. Search Student by Roll No
4. Update Student by Roll No
5. Delete Student by Roll No
6. Exit

Enter your choice: 1

Enter Roll No: 101

Enter Name: Riya Shah

Enter Gender: Female

Enter Contact: 9876543210

Enter Year: 2

Enter Branch: IT

 Student added successfully!

Enter Roll No to search: 101

Student Found:

Roll No: 101

Name: Riya Shah

Gender: Female

Contact: 9876543210

Year: 2

Branch: IT

Enter Roll No to update: 101

Enter New Name: Riya Patel

Enter New Gender: Female

Enter New Contact: 9988776655

Enter New Year: 3

Enter New Branch: CSE

 Student record updated successfully.

Enter Roll No to delete: 101

 Student deleted successfully.

